

CLASS :
EXAM NO :

TIME : 3 Hrs
MARKS : 80

SECTION A (1 mark each) (14 × 1 = 14)

1) Which of the following is a characteristic feature of phylum Ctenophora?

- a) Presence of comb plates for locomotion
- b) Presence of cnidoblasts
- c) Presence of water vascular system
- d) Presence of jointed appendages

2) Sea walnuts are commonly known as:

- a) Ctenophores
- b) Coelenterates
- c) Platyhelminthes
- d) Echinoderms

3) Bioluminescence is a well-marked feature in:

- a) Ctenophores
- b) Platyhelminthes
- c) Annelids
- d) Arthropods

4) The level of organization exhibited by ctenophores is:

- a) Cellular level
- b) Tissue level
- c) Organ level
- d) Organ-system level

5) In ctenophores, digestion is:

- a) Only extracellular
- b) Only intracellular
- c) Both extracellular and intracellular
- d) Neither extracellular nor intracellular

6) Ctenophores are:

- a) Asymmetrical and diploblastic
- b) Radially symmetrical and diploblastic
- c) Bilaterally symmetrical and triploblastic
- d) Radially symmetrical and triploblastic

7) Which of the following is exclusively marine?

- a) Porifera
- b) Ctenophora
- c) Coelenterata
- d) Platyhelminthes

8) The locomotion in ctenophores is aided by:

- a) Parapodia
 - b) Tube feet
 - c) Comb plates
 - d) Cilia
- 9) Which of the following is NOT a characteristic feature of Ctenophora?
- a) Radial symmetry
 - b) Diploblastic organization
 - c) Presence of cnidoblasts
 - d) Presence of eight external rows of ciliated comb plates
- 10) Reproduction in ctenophores is:
- a) Asexual only
 - b) Sexual only
 - c) Both sexual and asexual
 - d) By fragmentation
- 11) In ctenophores, sexes are:
- a) Separate
 - b) Not separate (hermaphrodite)
 - c) Parthenogenetic
 - d) Exhibit alternation of generations
- 12) The body organization in ctenophores is:
- a) Cellular level
 - b) Tissue level
 - c) Organ level
 - d) Organ-system level
- 13) Pleurobrachia is an example of:
- a) Porifera
 - b) Coelenterata
 - c) Ctenophora
 - d) Platyhelminthes
- 14) Which of the following is a distinctive feature of ctenophores?
- a) Presence of notochord
 - b) Presence of water vascular system
 - c) Presence of comb plates for locomotion
 - d) Presence of flame cells

SECTION B (2 marks each) (10 × 2 = 20)

- 15) What are the distinctive features of the phylum Ctenophora?
- 16) How do ctenophores differ from cnidarians in their mode of locomotion?
- 17) How does digestion occur in ctenophores?
- 18) What is bioluminescence in ctenophores?
- 19) Why is the symmetry of ctenophores considered radial?
- 20) How do ctenophores reproduce?
- 21) What is the tissue level of organization in ctenophores?

- 22) What is the significance of comb plates in ctenophores?
- 23) How are ctenophores exclusively marine organisms adapted to their habitat?
- 24) What is the diploblastic nature of ctenophores and how does it affect their body structure?

SECTION C (3 marks each) (7 × 3 = 21)

- 25) What are the characteristic features of phylum Ctenophora and how do they differ from other marine organisms?
- 26) Describe the locomotion mechanism in ctenophores. How does this differ from other marine animals?
- 27) Explain the significance of bioluminescence in ctenophores. What advantage might this feature provide to these organisms?
- 28) Describe the body organization and symmetry seen in ctenophores. How does this relate to their evolutionary position?
- 29) Compare the digestive system of ctenophores with that of cnidarians, highlighting the similarities and differences.
- 30) Explain the reproductive strategy in ctenophores. Why is this significant from an evolutionary perspective?
- 31) How are ctenophores different from cnidarians in terms of their cellular organization and specialized structures?

SECTION D (4 marks each) (2 × 4 = 8)

- 32) A marine biologist discovered a new species with eight rows of ciliated comb plates and tissue level organization, but with bilateral symmetry during its larval stage. The organism exhibits bioluminescence and has both intracellular and extracellular digestion.
- a) Identify the phylum to which this organism belongs and explain your reasoning.
- b) How does the digestive process in this organism differ from that found in Porifera?
- c) What evolutionary significance might the presence of both radial symmetry in adults and bilateral symmetry in larvae have?
- d) Explain how the locomotion mechanism in this organism differs from that in cnidarians. (4 marks)
- 33) During a marine expedition, researchers collected several specimens of comb jellies from different ocean depths. They observed variations in bioluminescence intensity among these specimens.
- a) What is the biological significance of bioluminescence in ctenophores?
- b) Explain how the tissue level of organization in ctenophores represents an evolutionary advancement over cellular level organization.
- c) Compare and contrast the locomotion mechanisms of ctenophores with those of echinoderms.
- d) How might the reproductive strategy of ctenophores (with non-separate sexes) provide advantages in their marine habitat? (4 marks)

CASE STUDY 2: PLANT ANATOMY AND ADAPTATION

Read the following passage and answer the questions that follow:

The anatomy of monocot and dicot plants shows marked differences in their internal structures. They differ in type, number and location of vascular bundles. Monocot stems have scattered vascular bundles while dicots have a ring arrangement. Monocot roots typically have more than six (polyarch) xylem bundles compared to fewer in dicot roots. In leaves, the monocots have parallel venation reflected in the near similar sizes of vascular bundles, while dicots show reticulate venation with varying vascular bundle sizes. Some monocot leaves have specialized bulliform cells that help the leaves curl inwards during water stress to minimize water loss.

Questions:

SECTION E (5 marks each) ($3 \times 5 = 15$)

Long Answer (5 marks each) Questions with Internal Choice

1) Describe the distinctive features and examples of Phylum Ctenophora. Explain how they differ from other lower invertebrates in terms of their body organization and locomotion.

OR

Describe the characteristic features of Phylum Cnidaria (Coelenterata). Explain the significance of cnidoblasts and the phenomenon of metagenesis with suitable examples.

2) Explain the morphological and anatomical differences between monocotyledonous and dicotyledonous stems with the help of well-labeled diagrams. How do these differences reflect their evolutionary adaptations?

OR

Describe the internal structure of a dorsiventral leaf with the help of a well-labeled diagram. How does its anatomy correlate with its physiological functions? Explain the structural adaptations that help in the process of photosynthesis.

3) Describe the morphology and anatomy of the frog's digestive system. How is it adapted to its carnivorous mode of life?

OR

Explain the respiratory system of frog. How is the frog adapted for respiration both in water and on land? Describe the mechanism of gaseous exchange in frogs during different environmental conditions including hibernation and aestivation.

ANSWER KEY (SECTION A)

- Q1. (a)
- Q2. (a)
- Q3. (a)
- Q4. (b)
- Q5. (c)
- Q6. (b)
- Q7. (b)
- Q8. (c)
- Q9. (c)
- Q10. (b)
- Q11. (b)
- Q12. (b)
- Q13. (c)
- Q14. (c)